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FEDERAL SPECIFICATION

PIPETTES, MILK AND CREAM EXAMINATION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers miscellaneous pipettes suitable for use in milk and cream testing procedures.

1.2 Classification. — Pipettes covered by this specification shall be of the following types, as specified:

Type I.—Breed and brew pipette
Type II.—Milk pipette
Type III.—Cream pipette
Type IV.—Milk diluting pipette (1.1 ml.)
Type V.—Milk diluting pipette (11 ml.)

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 Specifications. — There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Copies of Federal Specifications and the Index of Federal Specifications and Standards may be obtained upon application accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, SW., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Prices may be obtained from the Index of Federal Specifications and Standards or from the GSA Regional Offices. Single copies of this Specification and other product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.)

2.2 Other publications. — The following publications, of the specific issues shown, form a part of this specification:

American Public Health Association:

Standard Methods for the Examination of Dairy Products, 8th Edition
Standard Methods for the Examination of Dairy Products, 9th Edition

(Copies of the above referenced publications may be obtained upon application to the American Public Health Association, 1790 Broadway, New York, N. Y.)

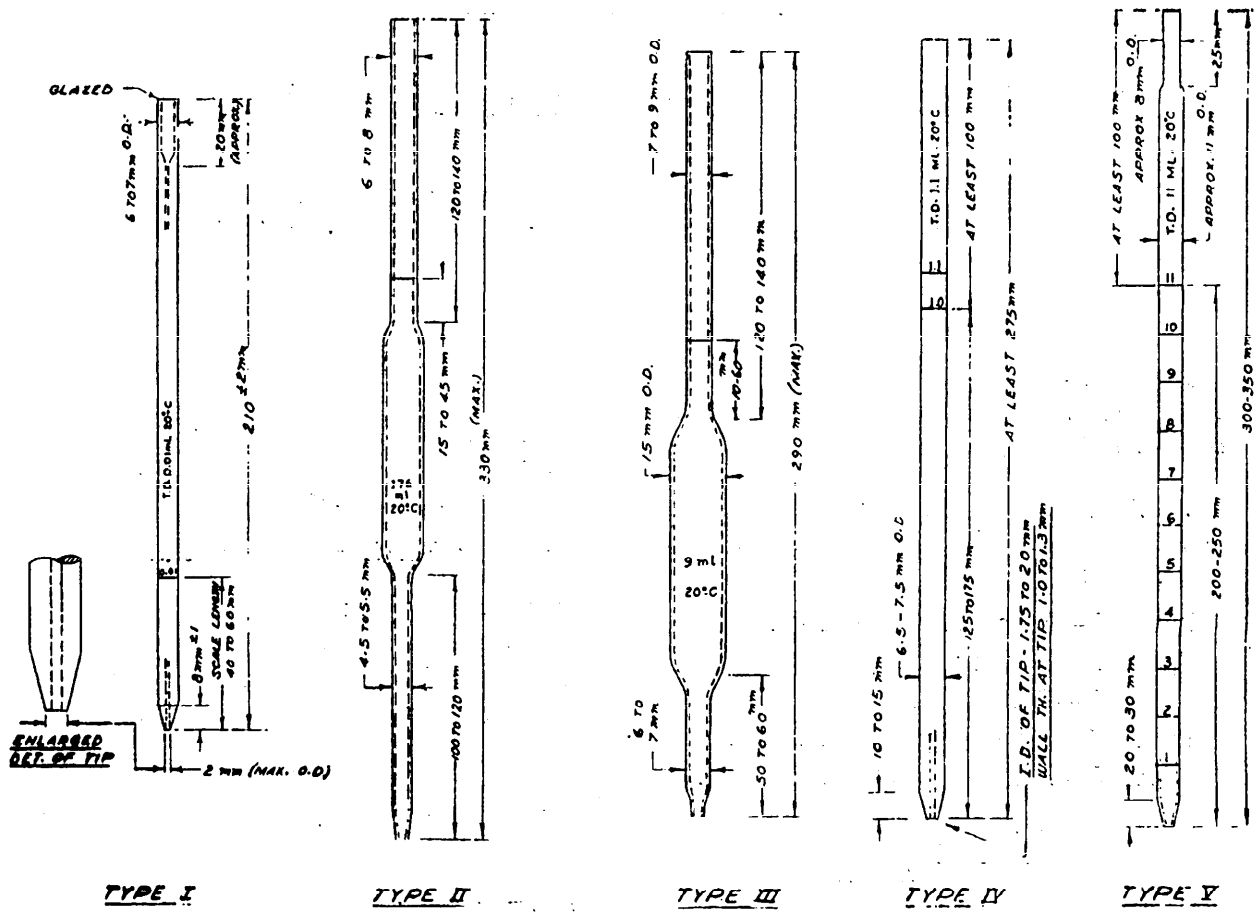
3. REQUIREMENTS

3.1 Material.—Material shall be high quality apparatus glass suitable in all respects for the purpose intended, and capable of passing the boiling test specified in 4.3.1.

3.2 Style and design.

3.2.1 *Type I, breed and brew pipette.*—Style and design of breed and brew capillary pipette shall be in strict conformance with figure 1. Pipette shall be without bulb, and shall be suitable for use in the direct count method of milk examination in conformance with the test procedures described in the 9th edition of Standard Methods for the Ex-

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Type	Nomenclature
I	Pipette, Capillary, Breed & Brew
II	Pipette, Milk
III	Pipette, Cream
IV	Pipette, Milk Diluting (1.1 ml.)
V	Pipette, Milk Diluting (11 ml.)

FIGURE 1.—*Pipettes, Milk and Cream Examination.*

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amination of Dairy Products, for pipette and slide method. The pipette shall be calibrated to deliver, when blown, 0.01 ± 0.001 ml. of milk having a specific gravity of 1.032 at 20°C. (equivalent to 0.1395 ± 0.0140 grams of mercury at 20°C.). Dimensions shall be as indicated on figure 1. Tips shall be blunt, ground, and polished to discharge milk cleanly.

3.2.2 *Type II, milk pipette.*—Milk pipette shall be a Babcock pipette suitable for testing the butter fat content of milk. The ends shall be at right angles to the axis. The bulb shall be symmetrical. The tip shall taper slightly over the terminal 5 to 20 mm., but the inside diameter at the end must be such as to deliver within 5 to 8 seconds. The end shall be fire-polished. Pipette shall be calibrated to contain 17.6 ± 0.05 ml. of water at 20°C., when bottom of meniscus coincides with mark on suction tube. Pipette shall be marked to indicate that it contains 17.6 ml. at 20°C. Pipette shall comply with the requirements shown in figure 1.

3.2.3 *Type III, cream pipette.* — Cream pipette shall be in accordance with figure 1, and calibrated to deliver 9 ± 0.10 ml. at 20°C., using distilled water, with drop remaining in tip blown out two seconds after delivery ceases and added to free delivery volume.

3.2.4 *Type IV, milk diluting pipette* (1.1 *ml.*)—Milk diluting pipette shall be in accordance with figure 1, of 1.1 ml. capacity, and graduated at 1.0 ml. and at 1.1 ml. to deliver. Pipette shall be calibrated to contain 1.075 and 1.180 ml. of distilled water at 20° C. at the 1.0 and 1.1 ml. graduations, respectively. Applicable tolerance shall be ± 0.025 ml. from 1.0 ml. to tip, from 1.1 ml. to tip, and from 1.1 to 1.0 ml.

3.2.5 *Type V, milk diluting pipette (11 ml.).*—Pipette shall be suitable for the dilution of milk during bacteriological examination, in accordance with the 8th edition of Standard Methods for the Examination of Dairy Products. Pipette shall be in accordance with figure 1. Capacity to deliver shall be 11 ml. graduated in 1 ml. divisions. Pipette shall be calibrated to contain 11.6 ± 0.10 ml. (1.055 ml. for each graduation interval) of distilled water at 20°C. from the 11 ml. graduation to tip. Tolerance of intermediate graduations to tip shall also be ± 0.10 ml.

3.3 Markings.—Graduations and capacity markings shall be deeply etched, distinct, and filled with fused-in color, and shall pass the tests for permanency outlined in 4.3.2. In addition, pipettes shall be permanently and legibly marked with the name or registered trade-mark of the manufacturer.

3.4 Workmanship.—Workmanship shall be first class throughout. Pipettes shall be free from defects which detract from their appearance or which may impair their serviceability. Pipettes shall be uniformly made, and top and bottom smoothly ground. Pipettes shall be properly annealed. When viewed in a polariscope, there shall be no ring strain; any longitudinal strain present shall be faint and highly diffused.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, at least six pipettes shall be taken at random from each lot of 100 or fraction thereof delivered, for inspection and tests.

4.2 Inspection. — Samples shall be inspected for material, style and design, markings, and workmanship.

4.3 Tests.

4.3.1 *Boiling test.*—Pipettes taken for test shall be completely immersed in distilled water in a closed vessel having a small steam vent, and boiled continuously for 6 hours, boiling water being added when necessary to compensate for evaporation losses. The pipettes shall be suspended in the water so

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that they do not come in contact with the containing vessel. As a result of this boiling test the pipettes shall show no evidence of corrosion, scumming, chipping, or cracking, and shall show no change in the calibrated volume requirements for each type specified herein.

4.3.2 *Permanent color of markings.* — In performing the following, a different sample shall be used for each test.

4.3.2.1 *Alkali test.*—Immerse samples completely in normal sodium hydroxide at room temperature, cover, and let stand 1½ hours. Remove samples and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

4.3.2.2 *Acid test (hydrochloric).*—Immerse samples completely in concentrated hydrochloric acid (sp. gr. 1.18) at room temperature, and let stand for 1½ hours. Remove samples and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

4.3.2.3 *Acid test (sulphuric).* — Immerse samples completely in concentrated sulphuric acid at room temperature. Let stand 1 hour. Remove samples and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, each pipette shall be packaged in accordance with accepted commercial practice.

5.2 Packing.—The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Packages.*—Unless otherwise specified, packages shall be marked with the name and size of the commodity, the quantity contained therein, and the name of the manufacturer and contractor.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name and size of the commodity as defined by the contract or order under which the shipment is made, the quantity contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 Purchasers should exercise any desired option offered herein. (See 1.2 and 7.2.3.)

6.2 This specification covers the following items appearing in the Armed Services Catalog of Medical Materiel:

4–365–325 Pipette, Capillary, Breed and Brew
4–365–350 Pipette, Cream
4–365–850 Pipette, Milk
4–365–860 Pipette, Milk Diluting
4–365–890 Pipette, Milk Diluting, 11 ml.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine conformance to the requirements of this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

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6.5 Transportation description.

6.5.1 *Transportation description applicable to these items is:*

Glassware, Chemical Laboratory.
Carload weight 18,000 pounds subject to Rule, 34, Uniform Freight Classification.
Truckload weight 18,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standard.*

Federal Specifications:

NN-B-621—Boxes, Wood, Nailed and Lock-Corner.
LLL-B-631—Boxes; Fiber, Corrugated (for Domestic Shipment).

Military Specifications:

JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN-P-133—Packaging and Packing for Overseas Shipment—Boxes, Set-Up, Paperboard.
MIL-L-10547 — Liners, Case, Waterproof.

Military Standard:

MIL-STD-129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

7.2.2 *Packaging.*

7.2.2.1 *Unit packaging.*—One pipette constitutes one unit. Each unit shall be packaged in a spiral wound paperboard tube or in a set-up paperboard box of appropriate size and design. Carton or tube shall be adequately secured to prevent accidental opening.

7.2.2.2 *Intermediate packaging.* — Twelve unit packages (12 units) shall be packaged in a carton of appropriate size conforming to the requirements of Military Specification JAN-P-133 or Federal Specification LLL-B-631. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Each carton shall be adequately secured to prevent accidental opening.

7.2.3 *Packing.*

7.2.3.1 *For domestic shipment.*—The number of intermediate cartons specified in table I shall be overpacked in an exterior container designed for type 3 load. Construction, closure, and strapping (when required) shall be in accordance with the requirements of Federal Specification NN-B-621.

7.2.3.2 *For overseas shipment.*—The number of intermediate cartons specified in table I shall be overpacked in an exterior container designed for type 3 load. Construction, closure, and strapping shall be in accordance with the requirements of Military Specification JAN-P-106.

7.2.3.2.1 *Case liner.*—Each box shall be

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lined with a waterproof case liner conforming to the requirements of Military Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

TABLE I.— *Packing quantities*

Stock number	Number of intermediate cartons to be contained in each shipping container	Total number of units to be contained in each shipping container
4-365-325	48	576
4-365-350	36	432
4-365-850	36	432
4-365-860	48	576
4-365-890	24	288

7.2.3.3 *Packing variation permitted.* — If the number of units in any shipment is less than the specified number of units to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design which will insure safe delivery to destination.

7.2.4 *Marking.*

7.2.4.1 *Interior packages.*—Each unit package (tube or carton) and each intermediate carton shall be marked with stock number, nomenclature, unit, quantity, office identifying number, and name of contractor.

7.2.4.2 *Shipping containers.* — Shipping container shall be marked in accordance with the requirements of Military Standard MIL-STD-129.

7.2.5 *Ordering data.* — Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packed for domestic or overseas shipment. (See 7.2.3.1 and 7.2.3.2.)

MILITARY INTERESTS:

Army—M
Navy—Md S
Air Force

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